30. Exam Practice: Mixed Review

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Four operations & fractions:

Calculate 3/5 of 120, then add 25.

Answer: 3/5 of 120 = 72; 72 + 25 = **97**.

Example 2 — Ratio & percentages:

A sum of £90 is shared in the ratio 2:7. Find 10% of the smaller share.

Answer: Smaller share = 2/9 × £90 = £20; 10% of £20 = **£2**.

Example 3 — Speed, distance & time:

A cyclist travels 42 km in 1 hour 45 minutes. Find her average speed in km/h.

Answer: 1 h 45 min = 1.75 h; speed = 42 ÷ 1.75 = **24 km/h**.

Example 4 — Area & money:

A room measures 7m by 4m. Carpet costs £6 per square metre. What is the total cost of carpeting the room?

Answer: Area = 7 × 4 = 28 m². Cost = 28 × £6 = **£168**.

Example 5 — Multi-step algebra:

A number is doubled, then 9 is added, giving 31. What was the original number?

Answer: 2 × number + 9 = 31, so 2 × number = 22, so the number = 22 ÷ 2 = **11**.

Foundation

15 questions

1. Write the value of the digit 5 in the number 52,839.

2. Calculate 2,963 + 4,178.

3. Calculate 8,002 − 3,459.

4. What is 27 × 6? A) 152 B) 162 C) 172 D) 182

5. Calculate 4,536 ÷ 6.

6. What is 1/4 of 48?

7. Calculate 6.4 + 3.75. A) 9.9 B) 10.15 C) 10.9 D) 11.15

8. Find 10% of 160.

9. Share £50 in the ratio 1:4. What is the larger share?

10. If x + 7 = 18, what is the value of x? A) 9 B) 10 C) 11 D) 12

11. How many degrees are there in a straight line?

12. Convert 1.5 kg into grams.

13. The mean of 3, 5, 7 and 9 is: A) 5 B) 6 C) 7 D) 8

14. Select ALL that apply: which of these numbers are multiples of 6? A) 12 B) 18 C) 25 D) 30 E) 35

15. A car travels at a steady 30 mph for 3 hours. How far does it travel?

Standard

20 questions

1. Mia buys 2 notebooks at £1.65 each and a pen for 75p. How much does she spend in total?

2. A recipe needs 200g of flour to make 8 cookies. How much flour is needed for 20 cookies?

3. A train journey covers 150 miles at an average speed of 50 mph. How long does the journey take? A) 2 hours B) 2.5 hours C) 3 hours D) 3.5 hours E) 4 hours

4. In a class of 25 pupils, 20% walk to school. How many pupils walk to school?

5. A rectangular garden is 9m long and 4m wide. What is its area?

6. Sarah has £30. She spends 3/10 of it on a book. How much money does she have left?

7. A shop reduces a £60 jacket by 25% in a sale. What is the sale price? A) £42 B) £43 C) £44 D) £45 E) £46

8. Four friends share a £48 restaurant bill equally. How much does each person pay?

9. A jug holds 2 litres of juice. 500ml is poured out. How much juice remains, in millilitres?

10. Select ALL that apply: a bag contains red, blue and green counters in the ratio 2:3:5. Which statements are true? A) For every 10 counters, 2 are red. B) Green counters are the most common colour. C) Blue counters make up 3/10 of the bag. D) Red counters make up 1/5 of the bag. E) There are more red counters than green counters.

11. A clock shows 9:50am. What time will it show in 65 minutes?

12. A football match lasts 90 minutes plus 3 minutes of added time. If kick-off is at 2:00pm, at what time does the match end (ignoring any half-time break)?

13. A number is doubled and then 5 is subtracted, giving 27. What was the original number? A) 14 B) 15 C) 16 D) 17

14. A hall measures 12m by 10m. Carpet tiles cost £7 per square metre. What is the total cost of carpeting the hall?

15. A car's fuel tank holds 40 litres. The car uses fuel at a rate of 10 litres per 100km. How far can it travel on a full tank?

16. At a school fair, cupcakes are sold for £3 each and raffle tickets for £2 each. 30 cupcakes and 50 raffle tickets are sold. How much money is raised in total?

17. A tank is 4/5 full of water. It holds 50 litres when full. How many litres need to be added to fill it completely?

18. A score is increased by 10% as a bonus, then reduced by 10% as a penalty. The final score is 297. What was the original score, before any changes?

19. Dan, Eve and Finn share £90 in the ratio 1:2:3. How much more does Finn receive than Dan?

20. A cyclist rides for 2 hours at 12 km/h, then rides for 1 hour at 6 km/h. What is her average speed for the whole journey?

Hard

5 questions

1. A rectangle measures 9cm by 7cm. A square is made using exactly the same length of wire as the rectangle's perimeter. What is the area of this square?

2. A candle is 90mm tall. Every hour it burns down by an amount equal to a third of its original height. How tall is the candle after 2 hours, in mm?

3. Jamie adds together three even numbers that follow one after another (like 4, 6, 8) and gets a total of 90. What is the largest of his three numbers?

4. A scale model uses 1cm to represent 150m in real life. A road on the model measures 8cm long. How long is the real road, in metres?

5. A paddling pool is filled using a hose that pours in 400 litres per minute. After how many minutes will the pool contain 12,000 litres?

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